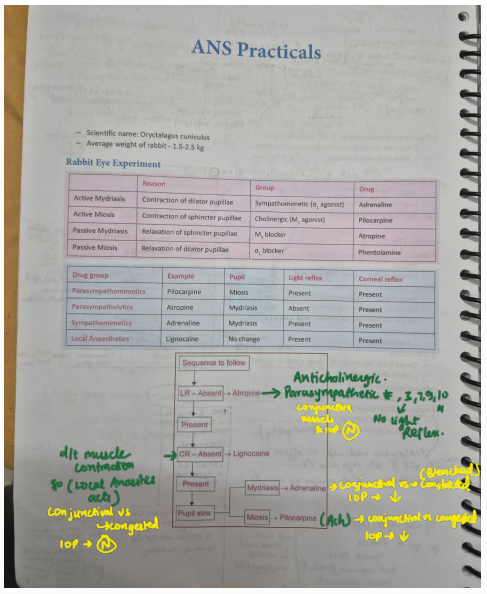
Light Reflex

Pupillary Reflex

*Corneal Reflex. → Mainly ⊖ by Local Anaesthetics only.

① Parasympatholytics
② " mimmetics
③ Sympathomimmetics

Drug Class	Pupil Size	Light Reflex	Corneal Reflex	Conjunctival Vessels	IOP
Anticholinergic (e.g., Atropine)	↑ (Mydriasis)	**Abolished**	Present	Normal	Normal
Sympathomimetic (e.g., Epinephrine)	↑ (Mydriasis)	**Present**	Present	Blenched (Constricted)	Reduced
Cholinergic (e.g., Pilocarpine)	↓ (Miosis)	Present	Present	Congested	Reduced
Local Anesthetic (e.g., Lignocaine)	Normal	Present	**Abolished**	Congested	Normal
Cocaine (Sympathomimetic + LA)	↑ (Mydriasis)	Present	**Abolished**	Blenched (Constricted)	Normal

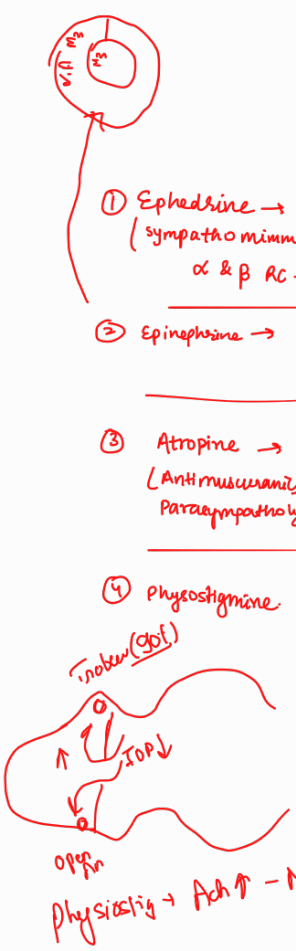


if not → Anticholinergic drug

Still having light reflex
↓
Pupillary diameter

N = 8mm.

	Pupil Size	Light Reflex	CORNEAL REFLEX	(14) IOT
① Ephedrine → mydriasis (↑) (sympathomimetic) α & β AC. stimulates.	9.5mm	+nt	✓	N
② Epinephrine →	↑ 7.5mm → 11mm	+nt		
③ Atropine → (Antimuscuranic) Parasympatholytic.	↑	✗	✓	N
④ Physostigmine	↓	✓	✓	↓ 9.8
⑤ Lignocaine	N	✓	✗	N
⑥ Phenylephrine	↑ 10mm	✓	✓	13.72



Sympathomimmetics → Indirectly < cocaine, TCA
TEPA
Directly
Catecholamines → Dopamine, Adrenaline
- Tyramine
- Ephedrine
- Pseudoephedrine
- Amphetamines.

14

N = 7

	Pupil Size	Light Reflex	Corneal Reflex	IOT
① Ephedrine → mydriasis (4) (sympathomimetic) Displacement of NA → NA↑	α & β RC stimulates. 8.5mm	+nt	✓	N
② Epinephrine → ↑ 7.5mm → 11mm		+nt	✓	12.46 ↓
③ Atropine → ↑ (Antimuscarinic) Parasympatholytic.		X	✓	N
④ Physostigmine ↓		✓	✓	↓ 9.8

inobtain (gpf)
↑ IOP ↓
open
physiolig → Ach↑ – M3 (+)

⑤ Lignocaine	N	✓	X	N
⑥ Phenylephrine	↑ 10mm	✓	✓	13.72
⑦ Pilocarpine ↓ 2mm (Direct-acting parasympathomimetic. M3 +)		✓	✓	11.2 ↓
⑧ Cocaine ↑ 9mm → Local Anesthetic as well as potent sympathomimmetic.		✓	X	N
⑨ Amphetamines ↑ 9mm		✓	✓	N

Rabbit Eye Experiment

An unknown drug solution was provided to you to know its nature through rabbit eye experiment. Following is the observation table:

Parameter	Control eye	Test eye
Pupil size	Normal	Increased
Light reflex	Present	Abolished
Corneal reflex	Present	Present
State of conjuctival blood vessels	Normal (Pinkish)	Normal (Pinkish)
Intraocular pressure	Normal	Normal

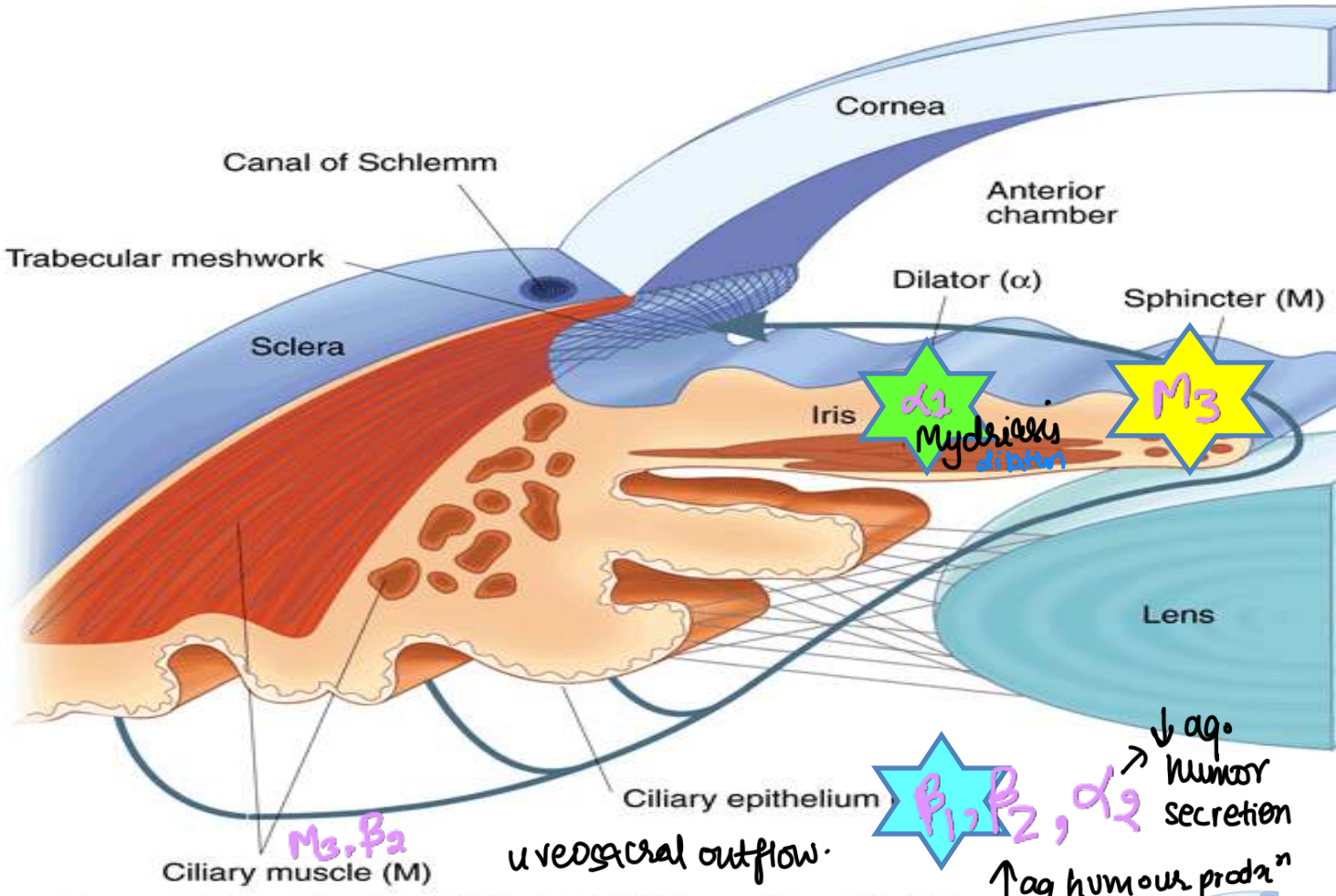
Anticholinergic ⇒ Eg Atropine

Q.1 Comment on the nature of unknown drug.

Q.2 Mention two important uses of drugs under this category.

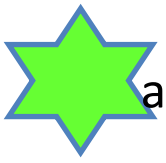
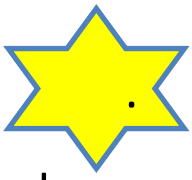
1. Scopolamine ⇒ Motion Sickness
2. Atropine ⇒ of poisoning, Fundoscopy, Refraction testing
3. Benzhexol, Benztropine ⇒ In Parkinsonism
4. Ipratropium, tiotropium ⇒ In Asthma, COPD
5. Solefenacin, oxybutynin ⇒ overactive bladder

M_1
M_2
M_3 ← Bronchus, Bladder, Glands, Eye, CNS.



Source: Katzung BG, Masters SB, Trevor AJ: *Basic & Clinical Pharmacology, 12th edition*: www.accessmedicine.com

Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

Q.1 Name the predominant receptors present on locations  and  .

Q.2 Name one anti-glaucoma drug acting predominantly by increasing trabecular outflow and one acting predominantly by increasing uveoscleral outflow of aqueous humor.

↳ ↑ Trabecular outflow ⇒ Pilocarpine, physostigmine

↳ ↑ uveoscleral outflow ⇒ Latanoprost, Bimatoprost

(PGF$_{2\alpha}$) ↳ Lato 2T Bimato

An unknown drug solution was provided to you to know its nature through rabbit eye experiment. Following is the observation table:

Parameter	Control eye	Test eye
Pupil size	Normal	Increased
Light reflex	Present	Present
Corneal reflex	Present	Present
State of conjuctival blood vessels	Normal (Pinkish)	Blenched
Intraocular pressure (IOP)	Normal	Decreased

Diff

Sympathomimetic ⇒ Epinephrine

Q.1 Comment on the nature of unknown drug.

Q.2 Mention the major mechanism of reduction in IOP by the unknown drug.

$\alpha_1 \Rightarrow$ mydriasis
$\alpha_2 \Rightarrow$ ↓s aqueous humor secretn
β_2 (ciliary epith.) ⇒ ↑s aqus humor productn
β_2 (ciliary muscle) ⇒ ↑se uveoscleral outflow
} ↓s IOP

Q.3 Mention two important uses of this category of drugs.

1. Adrenaline ⇒ Anaphylaxis, Cardiac arrest
2. Dobutamine ⇒ CHF
3. Apraclonidine ⇒ OAG
4. Ephedrine, Phenylephrine ⇒ nasal decongestant
5. Salbutamol, formetrol, terbutaline ⇒ Asthma
6. Amphetamine ⇒ CNS stimulants
7. Ritodrine, ⇒ Uterine relaxation

An unknown drug solution was provided to you to know its nature through rabbit eye experiment. Following is the observation table:

Parameter	Control eye	Test eye
Pupil size	Normal	Decreased
Light reflex	Present	Present
Corneal reflex	Present	Present
State of conjuctival blood vessels	Normal (Pinkish)	Congested
Intraocular pressure (IOP)	Normal	Reduced

⇒ Cholinogic

Q.1 Comment on the nature of unknown drug obtained from a plant as an alkaloid.

Q.2 Mention the probable mechanism of action of reduction in IOP by unknown drug. ⊕ M_3 receptors of ciliary muscles ⇒ Ciliary muscle Contraction ⇒ Pores are widen & Used trabecular outflow ⇓ ↓IOP

Q.3 Mention two important uses of drugs under this category.

1. Physostigmine ⇒ Atropine poisoning, AC G
2. Donepezil, Galantamine ⇒ Alzheimer's disease
3. Neostigmine + Atropine ⇒ myasthenia gravis, Cobra bite
4. Neostigmine ⇒ Post op Paralytic Ileus, Post op Urinary retension
5. Neostigmine + Atropine ⇒ Reversal of Action of non-depolarising muscle relaxants

An unknown drug solution was provided to you to know its nature through rabbit eye experiment. Following is the observation table:

Parameter	**Control eye**	**Test eye**
Pupil size	Normal	Normal
Light reflex	Present	Present
Corneal reflex	Present	Abolished
State of conjuctival blood vessels	Normal/Pinkish	Congested
Intraocular pressure	Normal	Normal

Amide Local anesthetic agent

Q.1 Comment on the nature of unknown drug.

Q.2 Mention two important adverse effects of drugs under this category.

① Lignocaine, Prilocaine (max), Benzocaine ⇒ Methemoglobinemia

② All LAs ⇒ (high dose) ⇒ Seizures followed by Coma

③ Bupivacaine ⇒ maximum Cardiotoxic

④ only LA causing VC ⇒ Cocaine

An unknown drug solution was provided to you to know its nature through rabbit eye experiment. Following is the observation table:

Parameter	**Control eye**	**Test eye**
Pupil size	Normal	Increased
Light reflex	Present	Intact
Corneal reflex	Present	Abolished
State of conjuctival blood vessels	Normal (Pinkish)	Blenched
Intraocular pressure (IOP)	Normal	Normal

Q.1 Name the unknown drug (alkaloid) which is obtained from a plant. Cocaine ⇒ Extracted from leaves of "Erythroxylem Coca"

An unknown drug solution was provided to you to know its nature through rabbit eye experiment. Following is the observation table:

Parameter	**Control eye**	**Test eye**
Pupil size	Normal	Decreased
Light reflex	Present	Intact
Corneal reflex	Present	Intact
State of conjuctival blood vessels	Normal (Pinkish)	Normal (Pinkish)
Intraocular pressure (IOP)	Normal	Reduced

Physostigmine

Q.1 Comment on the nature of unknown drug.

Q.2 Mention the probable mechanism of action of reduction in IOP by the unknown drug.

Q.3 Mention two important uses of drugs under this category.